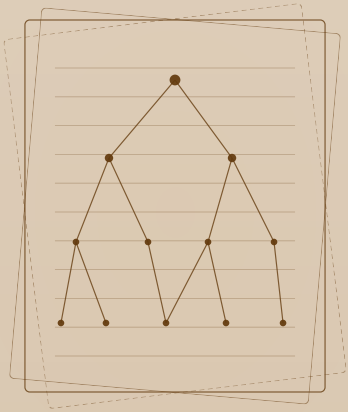
Ghosts in the Parchment



How Scholars Exposed Forgeries, Deciphered Lost Scripts, and Reconstructed Vanished Books

PHILOLOGY & FORENSICS

Ghosts in the Parchment

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Ghosts in the Parchment

The Forensic Science of the Written Word

Every surviving book from the ancient or medieval world is a physical machine. Over centuries, each text has endured mechanical wear, copying distortion, ideological censorship, and deliberate counterfeit. Before the invention of movable metal type, a work could outlive its organic reed or animal skin only if human scribes copied it by hand across generations. With each transcription, tired eyes skipped lines and unfamiliar foreign words mutated into local slang. Marginal glosses migrated into the main column, while political authorities inserted forged clauses to legitimate empires. Where copying stopped altogether, entire civilizations vanished behind unreadable scripts carved into limestone cliffs or pressed into wet clay.

For most of history, readers treated written documents either as sacred authorities beyond question or as opaque relics whose lost meanings had to be guessed by intuition. Beginning in the fifteenth century, a succession of philologists, cryptanalysts, statisticians, and physicists transformed textual study into a forensic science. Instead of asking what a book ought to mean, they treated every manuscript, inscription, and printed folio as a physical crime scene. Each artifact is governed by quantifiable laws of linguistic mutation, genealogical error propagation, and material chemistry. When an investigator measures the physical traces left behind in clay, papyrus, lead type, or scraped goatskin, erased voices speak again.

CHAPTER 1

The Anachronistic Words

Lorenzo Valla and the Forgery of the Donation of Constantine

Inside the royal chancery of Castel Nuovo in Naples in 1440, humanist Lorenzo Valla examined a Latin parchment charter that had shaped European geopolitics for six hundred years. While the armies of Pope Eugenius IV and King Alfonso V of Aragon fought for southern Italy, Valla audited the *Constitutum Constantini*, known as the *Donation of Constantine*. The document purported to be an imperial decree issued on March 30, 315 CE, under Emperor Constantine the Great and the consul Gallicanus. According to its text, Pope Sylvester I had cured Constantine of leprosy in Rome. In gratitude, the emperor handed over his imperial crown, purple cloak, and temporal jurisdiction over Rome, Italy, and the entire western Roman Empire to the bishop of Rome forever.

Throughout the Middle Ages, canon lawyers incorporated the *Donation* into Gratian's twelfth-century *Decretum*. Popes cited it to claim feudal supremacy over European kings. Earlier critics, including Otto of Freising and Nicholas of Cusa, had questioned whether any Roman emperor would voluntarily strip his dynasty of half the world. They also wondered why fourth-century chroniclers like Eusebius never mentioned the transfer. Yet political skepticism could not overturn a sealed charter in court. Serving as royal secretary to King Alfonso, Valla saw that the charter could be demolished from within. Instead of debating theology, he audited the physical medium of the document: its Latin grammar, institutional titles, and historical vocabulary.

Born in Rome to a family of curial lawyers, Valla had spent years compiling his six-volume *Elegantiae linguae Latinae*. That empir-

ical survey compared classical Latin syntax against medieval scholastic usage. Valla recognized that human language is not a static code. It mutates continuously over time, acquiring loanwords, shifting idioms, and coining terms tied to specific offices. A genuine constitution drafted in 315 CE had to obey the linguistic and administrative boundary conditions of the early fourth century. Any word, title, or biblical citation that originated centuries later acted like a radioactive isotope proving the text was composed after that date.

In his 1440 treatise *De falso credita et ementita Constantini donatione declamatio*, Valla dissected the charter sentence by sentence. First, he noted its complete absence from fourth-century records. No gold or bronze copy existed, no Roman senate decree ratified the partition, and Roman consuls continued ruling Italy under Constantine's sons without interruption. Second, Valla cataloged dozens of linguistic anachronisms embedded in the forger's prose:

- **The Unbuilt Capital* (Constantinopolis)*: The charter granted authority over the patriarchal sees of Alexandria, Antioch, Jerusalem, and Constantinople. Valla pointed out that in 315 CE, the city on the Bosporus was still a Greek town named Byzantium. Constantine did not decide to rebuild it until 324 CE. He did not dedicate it as Constantinople until May 11, 330 CE, fifteen years after the charter's date.
- **The Persian Officials* (Satrapae)*: Opening its imperial salutation, the decree addressed the empire's satraps alongside the Roman senate. Valla demonstrated that no Roman emperor or legal digest ever called provincial governors satraps. That Persian title was borrowed by a medieval cleric reading the Latin Vulgate books of Daniel and Maccabees.
- **The Cloth Crown* (Diadema *vs.* Corona)*: *The forger wrote that Constantine placed a diadem of pure gold and precious stones*

upon Pope Sylvester's head. Valla proved from classical coins that in fourth-century Latin, a diadema *was never a solid metal crown* (corona*). It was a silk or linen headband fastened around the forehead.

- **Clerical Vestments and Barbarisms**: The text granted Roman clergy the right to wear linen horse-cloths (*mappulae*) and ecclesiastical sandals (*campagi*). It also filled its sentences with eighth-century Frankish chancery barbarisms such as *condecet* and citations from St. Jerome's Latin Vulgate Bible, translated in the 380s CE.

By cross-referencing these vocabulary strata, Valla pinned the forgery to the eighth century. Around 754 CE, clerics in the papal chancery fabricated the charter to persuade the Frankish king Pepin the Short to hand central Italy over to the papacy. Modern palaeographers have confirmed Valla's window, tracing the earliest surviving copy (*BnF Latin 2777*) to the Frankish abbey of Saint-Denis. After Ulrich von Hutten printed Valla's treatise in 1517, philological anachronism became the standard forensic test for historical documents.

CHAPTER 2

The Cliff of Three Tongues

Henry Rawlinson and the Cuneiform Trilingual of Behistun

Three hundred feet up a limestone precipice at Mount Behistun in western Persia, British cavalry officer Henry Creswicke Rawlinson dangled from ropes above a vertical drop in the summer of 1835. Before him, chiseled into a smoothed rock face twenty-five feet high and fifty feet wide, stood a massive bas-relief carved in 520 BCE by order of the Achaemenid emperor Darius the Great. The sculpture showed Darius placing his foot on the chest of the defeated usurper Gaumata while nine bound rebel kings stood roped together. Surrounding the figures were over a thousand lines of wedge-shaped cuneiform script arranged in three parallel columns. A coat of siliceous varnish had preserved every chisel stroke for twenty-three centuries.

Unlike Egyptian hieroglyphs—which Jean-François Champollion decoded in 1822 using the Greek text on the Rosetta Stone—Mesopotamian cuneiform had died out by the first century CE without leaving a single bilingual Greek or Latin inscription. For two millennia, travelers walking the mounds of Nineveh, Babylon, and Persepolis stared at clay tablets covered in bird-track wedges. Nobody knew whether the marks were letters, ideograms, or floral ornaments. In the seventeenth and eighteenth centuries, European visitors Pietro della Valle and Carsten Niebuhr copied short inscriptions from Persepolis. Their transcripts showed that the carvings always grouped three wedge systems together. The first was a simpler script of forty-two signs (Old Persian). The second was a middle script of over a hundred

signs (Elamite). The third was a script of over five hundred characters (Babylonian Akkadian).

In 1802, German teacher Georg Friedrich Grotefend in Göttingen made the first crack in the simplest script. Inspecting Niebuhr's copies of two Persepolis doorway inscriptions where diagonal wedges divided words, Grotefend noticed a recurring royal formula: `[Name A], [Title], [Title] of [Title]s, son of [Name B].` Comparing the Achaemenid kings recorded by Herodotus, Grotefend looked for father-and-son rulers whose names were roughly the same length. Moreover, the father of the first king did not carry the royal title himself. Cyrus and Cambyses failed because both names began with the same letter, whereas the cuneiform names began with different signs. That left Hystaspes (a non-king), his son Darius, and his grandson Xerxes. Grotefend slotted in a dozen phonetic values, but he lacked long texts and Iranian philology, so his work stalled.

Stationed at Kermanshah in 1835, Rawlinson realized that the Behistun cliff was the Rosetta Stone of Mesopotamia. Returning between 1835 and 1837 with ladders bridged across eighteen-inch ledges, Rawlinson copied the Old Persian columns into his notebooks. Because Darius's proclamation listed dozens of rebellious provinces and generals across the empire—Babylon, Assyria, Media, Armenia, and Bactria—Rawlinson possessed nearly a hundred proper names. He cross-checked their phonetic spellings against Greek historians, the Zoroastrian *Avesta* studied by Eugène Burnouf, and classical Sanskrit:

- **Slot-Matching the Royal Genealogy**: Applying the dynastic template to the opening lines of Behistun, Rawlinson isolated the Old Persian phoneme chains for Hystaspes (*Vi-i-sh-ta-a-s-pa*), Darius (*Da-a-ra-ya-va-u-sh*), and Xerxes (*Kha-sha-ya-a-r-sha*). That step recovered the 42-sign Old Persian semi-alpha-

betic syllabary, in which consonants carried inherent vowel affinities (*a*, *i*, or *u*).

- **Comparative Indo-Iranian Grammar**: Once proper names yielded the phonetic values of all forty-two signs, Rawlinson translated the vocabulary by matching Old Persian roots and case endings against Sanskrit and Avestan cognates. He proved that *khshayathiya* (king, source of modern *shah*) corresponded to Sanskrit *kshatriya*, and *bumi* (earth) matched Sanskrit *bhumi*.
- **Cracking Polyvalent Babylonian**: Returning to Behistun in 1844 and 1847, Rawlinson hired a Kurdish boy who wedged wooden pegs into rock crevices above a chasm. Swinging across on a rope cradle, the boy beat wet sheets of paper into the stone to lift molds of the 112-line Babylonian column. Comparing that column against his Old Persian translation, Rawlinson and Irish clergyman Edward Hincks proved that Babylonian Akkadian was a Semitic language akin to Hebrew and Arabic. Its logosyllabic script used wedge signs as both ideograms and polyvalent syllables (*kur*, *mat*, *shat*).

In May 1857, the Royal Asiatic Society in London sealed Rawlinson, Hincks, Julius Oppert, and William Henry Fox Talbot in separate rooms with an unpublished cuneiform prism of the Assyrian king Tiglath-Pileser I. When their four independent translations matched line for line, Mesopotamian philology passed its decisive blind test.

CHAPTER 3

The Vanished Archetype

Karl Lachmann and the Forensic Reconstruction of Lucretius

Fifty medieval manuscripts copied directly from a single surviving parent book possess no more independent evidentiary weight than that one parent alone. For four hundred years after the invention of the printing press, however, classical scholars editing Greek and Roman literature routinely violated that principle. Faced with dozens of contradictory manuscript copies of a classical author, Renaissance and eighteenth-century editors selected the commonest reading by majority vote among late Italian manuscripts. Whenever a verse sounded awkward, they invented conjectural repairs (*emendatio ope ingenii*). No editor had established a mathematical procedure for tracing how surviving copies descended from a lost original.

Between 1845 and 1850, at the University of Berlin, German philologist Karl Lachmann transformed textual criticism from an art of subjective taste into a deterministic forensic science. Working across classical Latin poets and medieval Middle High German epics, Lachmann split editorial work into two strictly separated stages. First came *recensio*, the mechanical reconstruction of the earliest recoverable ancestor exclusively from manuscript evidence. Second came *emendatio*, repairing corruptions already present in that ancestor. As his proving ground, Lachmann chose Titus Lucretius Carus's first-century BCE six-book epic *De Rerum Natura* (*On the Nature of Things*). That poem explained the universe through Epicurean physics, where indivisible atoms fall and swerve (*clinamen*) through infinite void.

During the Christian Middle Ages, Lucretius's materialist treatise had nearly vanished. For centuries, all printed editions descended from a single lost copy discovered in a German monastery in January 1417 by papal secretary Poggio Bracciolini, who mailed a transcript to Niccolò Niccoli in Florence. From Niccoli's copy sprang scores of fifteenth-century Italian humanist manuscripts filled with scribal interpolations. Instead of polling those late Italian copies, Lachmann focused on two ninth-century Carolingian parchment codices preserved at Leiden in the Netherlands. The first was the *Codex Oblongus* (O, Vossianus Latinus F 30), copied around 800 CE at Charlemagne's court and corrected by the Irish monk Dungal. The second was the *Codex Quadratus* (Q, Vossianus Latinus Q 94), copied in northern France later in the ninth century, alongside thirty-four ninth-century loose leaves known as the *Schedae* (S) in Copenhagen and Vienna.

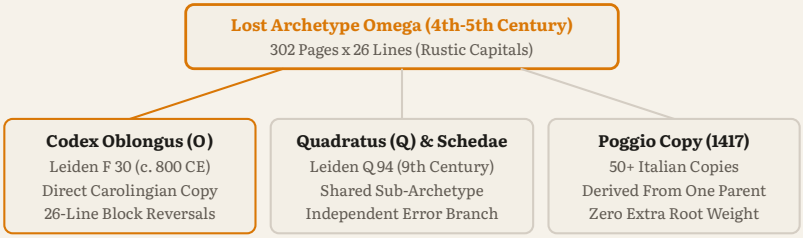


FIG. 3.1 — Karl Lachmann's stemmatic tree (stemma codicum) reconstructing the vanished 302-page Archetype of Lucretius from shared scribal errors.

To determine how O, Q, and Poggio's Italian family were related, Lachmann used shared scribal errors the way a geneticist uses shared DNA mutations. When a scribe copies a text accurately, agreement between two manuscripts tells nothing about their family tree because both preserve the author's original words.

When a scribe commits an unmistakable blunder—skipping a line due to identical word endings (*homoeoteleuton*) or leaving a gap where a parent page was torn—every descendant copied from that manuscript inherits the exact same wound. By cataloging conjunctive errors (*errores coniunctivi*) shared across all witnesses alongside separative errors (*errores separativi*) unique to O or the Q-S branch, Lachmann proved that all surviving copies descended from a single lost antique manuscript, the Archetype (Ω).

By auditing the arithmetic of displaced verses inside O and Q, Lachmann even reconstructed the physical pagination of that vanished fourth- or fifth-century archetype. In Book IV of *De Rerum Natura*, verses 323–347 and verses 299–322 appeared in reversed order. Those two transposed blocks contained twenty-five lines plus one heading line each, or 26 lines per block. Across the poem, whenever a leaf of the lost archetype had been torn out or pasted in backward, the resulting gap or transposition in the ninth-century copies obeyed a strict integer quantization law for the number of displaced lines $L_{\text{displaced}}$:

$$L_{\text{displaced}} = 26 \times k \quad (k \in \{1, 2, \ldots\})$$

Every leaf (*folium*) of the lost archetype carried 26 lines on its front (*recto*) and 26 lines on its back (*verso*), or 52 lines per sheet. Counting every verse and header across all six books, Lachmann proved in his 1850 Berlin edition that the vanished archetype had been a parchment codex of 302 written pages of 26 lines each, penned in rustic capitals without word spacing.

CHAPTER 4

The Sacred Attic

Solomon Schechter, the Twin Sisters, and the Cairo Geniza

A single torn leaf of brownish vellum covered in cramped medieval Hebrew script, bought from an antiquities dealer in Cairo by Scottish twin sisters Agnes Smith Lewis and Margaret Dunlop Gibson in the spring of 1896, carried a text unseen for nearly nine hundred years. Returning to Cambridge that May, the twins were already famous as self-taught polyglots fluent in twelve languages. Four years earlier, they had discovered the fourth-century *Sinaitic Palimpsest* of the Old Syriac Gospels at Saint Catherine's Monastery. On May 13, 1896, they walked down Trumpington Street and handed the parchment fragment across a table to Solomon Schechter, Reader in Talmudic and Rabbinic Literature at Cambridge University.

Schechter did not even need a lexicon. Within an hour of examining the faded ink in the Cambridge University Library, he sent a telegram to the sisters asking them to keep the find confidential. The fragment was a leaf from the original second-century BCE Hebrew book of *The Wisdom of Jesus Ben Sira*, known in Christian Bibles as *Ecclesiasticus*. Written in Jerusalem around 180 BCE by the Jewish scribe Shimon ben Yeshua ben Eleazar ben Sira, the book had been translated into Greek by the author's grandson in Egypt in 132 BCE. It was preserved in the Greek Septuagint and Latin Vulgate canons, but excluded from the rabbinic Hebrew Bible. Although tenth-century Babylonian scholar Saadya Gaon had quoted the Hebrew original, every Hebrew manuscript had vanished after the tenth century. Schechter now held chapters 39 and 40 in his hands.

Where in Egypt could a tenth-century Hebrew parchment survive unrotted for a millennium? Across Europe, humid soils and damp monastery cellars decompose organic rag paper and animal skins within a few centuries. Expulsions and routine vellum recycling also destroyed the everyday letters and contracts of medieval Jewish communities. In Fustat (Old Cairo), however, two forces converged to create a time capsule: the rainless desert climate and a strict rule of Jewish ritual law (*halakha*). Jewish law forbids destroying any piece of parchment or paper bearing the written name of God (*shemot*). Because medieval Hebrew letters, marriage contracts, and business ledgers routinely opened with pious invocations of God, residents refused to throw worn-out writing into the street trash. Whenever books tore or correspondence piled up, householders climbed a ladder inside the Ben Ezra Synagogue (rebuilt in 1040 CE). They dropped the discarded sheets through a high wall opening into a dark, windowless two-story shaft called the *geniza* (storeroom).

Financed secretly by the Master of St. John's College, Charles Taylor, Schechter sailed for Egypt in December 1896. After winning the trust of Cairo's Chief Rabbi, Aaron Raphael Ben-Shim'on, Schechter was shown the ladder leading up to the sealed chamber. Climbing into a windowless room where centuries of dry masonry dust rose in choking clouds, Schechter encountered nearly three hundred thousand leaves of vellum, rag paper, and papyrus spanning ten centuries from 870 CE to the 1880s:

- **Lost Biblical and Apocryphal Texts**: Alongside further leaves of the Hebrew *Ben Sira*, Schechter uncovered sixth-century palimpsests carrying Aquila's second-century Greek translation of the Bible. He also found two medieval manuscripts of the *Damascus Document*, a sectarian Jewish rulebook whose first-

century BCE ancestors were unearthed fifty-one years later among the Dead Sea Scrolls at Qumran.

- **Autographs of Medieval Philosophers and Poets**: Buried in the dust lay dozens of pages written in the personal hand of the twelfth-century physician and philosopher Moses Maimonides. These included early rough drafts of his *Mishneh Torah* and Arabic medical responsa bearing his own revisions, plus lost poems by Judah Halevi.
- **The Mediterranean Commercial Ledger**: Unlike royal archives that saved only state treaties, the Ben Ezra Geniza preserved unfiltered daily paperwork later cataloged by historian S. D. Goitein. Its divorce decrees, trousseau inventories, pharmacy prescriptions, and eleventh- and twelfth-century Judeo-Arabic merchant letters tracked pepper, flax, silk, and bronze shipments across Tunisia, Sicily, Aden, and India.

Working through clouds of dust inside the Cairo chamber, Schechter packed crates holding 193,000 fragments for Cambridge. His haul restored the lost Hebrew original of Ecclesiasticus, handwritten drafts by Maimonides, and twelfth-century trade invoices linking Egypt to the Malabar coast.

CHAPTER 5

The Coffeehouse Homer

Milman Parry, Albert Lord, and the Oral Machinery of Epic

Composing twelve thousand lines of strict dactylic hexameter verse aloud at the speed of four syllables per second without a single written note defies ordinary human working memory. A literate poet sitting at a desk with a pen can pause for twenty minutes to hunt for a word whose long and short vowel quantities fit the meter (– u u | – u u | – u u | – u u | – u u | – x). The writer can cross out a clumsy line and revise an earlier stanza at leisure. An improvising oral singer performing live before a crowded hall has zero seconds to hesitate. If the singer stops to calculate syllable lengths, the rhythm collapses.

For more than a century after German philologist Friedrich August Wolf published his *Prolegomena ad Homerum* in 1795—arguing that writing was largely unknown in eighth-century BCE Greece when the *Iliad* and *Odyssey* took shape—classical scholarship split into two camps over the Homeric Question. The Analysts dissected the epics into separate ballads stitched together by later editors. They pointed to narrative contradictions and the mechanical repetition of stock adjectives. Why did Homer call Achilles swift-footed (*podas okys*) while the hero sat still inside his tent, or call the villain Aegisthus blameless? The Unitarians countered that the architectural symmetry of the *Iliad* proved it was the work of a single literate genius who used epithets as stylistic ornament.

In 1928, American classicist Milman Parry demolished both camps at once in two French doctoral theses at the Sorbonne in Paris, led by *L'Épithète traditionnelle dans Homère*. Instead of treating swift-footed Achilles or much-enduring divine Odysseus as literary

decorations, Parry tabulated every noun-epithet phrase across all 27,803 lines of the *Iliad* and *Odyssey*. He sorted each formula by the grammatical case of the hero's name and the exact metrical shape of the space remaining between the line's mid-verse pause (*caesura*) and the end of the hexameter verse.

When Parry laid those tables side by side, a rigorous algebraic structure emerged. The Homeric epithet system obeyed two iron laws that Parry named **extension** and **economy** (or **thrift**):

- **Fixed Metrical Slot Coverage (Extension)**: In the second half of a dactylic hexameter line, a poet needing to name Odysseus in the nominative case faces three standard metrical gaps depending on where the first half-line ended. The break falls after the trochaic caesura in the third foot, the penthemimeral caesura in the third foot, or the hephthemimeral caesura in the fourth foot. For Odysseus, the tradition supplies three phrases of descending length: *polytlas dios Odysseus* (much-enduring divine Odysseus, - u u - - - x), *polymētis Odysseus* (many-minded Odysseus, u u - u u - x), and *dios Odysseus* (divine Odysseus, - u u - x).
- **Strict Non-Redundancy (Thrift)**: Across all thirty-seven major characters in the *Iliad* and *Odyssey*, for any given grammatical case and metrical slot length, there is almost never a second competing noun-epithet formula with the same metrical shape. Out of dozens of possible adjectives meaning brave or noble, the tradition kept one per metrical slot and discarded all duplicates.
- **Acoustic and Syntactic Automation**: Because the choice of epithet was dictated by the grammatical case of the verb and the metrical boundary of the preceding clause—never by what the character was physically doing—an oral poet could build the first half of a verse freely. The singer then snapped the

pre-measured epithet block into the second half automatically while planning the next line.

No single literate author inventing phrases at a desk writes with that algebraic thrift. Such a system could only be forged over generations by pre-literate oral singers. Between 1933 and 1935, accompanied by his student Albert Bates Lord and local bard Nikola Vujnović, Parry traveled into the Dinaric Alps of Yugoslavia. There illiterate Bosnian and Montenegrin bards (*guslari*) still sang heroic epics in mountain coffeehouses while bowing a single-stringed *gusle*.

To record twelve-hour songs without stopping a singer every four minutes, Parry commissioned a battery-powered dual-turntable cutting rig that toggled continuously between twelve-inch aluminum discs. In Bijelo Polje in July 1935, the illiterate butcher Avdo Međedović sang *The Wedding of Smailagić Meho* across five days onto Parry's turntables, generating 12,311 epic lines equal in length and complexity to Homer's *Odyssey*.

CHAPTER 6

The Inflectional Grid

Alice Kober, Michael Ventris, and the Decipherment of Linear B

Stacked inside empty cigarette cartons and biscuit tins in a Flatbush, Brooklyn apartment between 1943 and 1950, 180,000 two-by-three-inch index cards cut from wartime scrap paper held the internal grammar of Bronze Age Crete. Their creator, Brooklyn College classics professor Alice Elizabeth Kober, carried a full teaching load by day and translated textbooks into Braille for blind students. At night she taught herself Hittite, Akkadian, Sanskrit, Basque, and Old Irish to attack a script that had defied every cryptanalyst on Earth for half a century.

In the spring of 1900, British archaeologist Sir Arthur Evans dug into the Bronze Age palace of Knossos on Crete and uncovered more than three thousand clay tablets dated to around 1400 BCE. Scratched into the clay were inventories of sheep, chariots, bronze cauldrons, and workers written in an unknown script that Evans named **Linear B**. In 1939, American archaeologist Carl Blegen unearthed six hundred more Linear B tablets inside the Mycenaean palace at Pylos on the Greek mainland. Yet Evans, who controlled access to most of the Knossos tablets until his death in 1941, enforced a strict dogma. He insisted that the Minoans of Crete had colonized mainland Greece and that Linear B recorded an indigenous Minoan language that could not be Greek.

To a cryptanalyst, Linear B represented the hardest class of problem in information theory: an unknown script writing an unknown language with zero bilingual inscriptions. Worse, the tablets were terse accounting lists containing no royal genealogies or narrative prose. Kober realized that guessing phonetic values

by comparing Linear B shapes to Cypriot or Phoenician letters was futile. Before anyone could ask how a sign sounded, one had to map how the signs behaved mathematically in relation to one another.

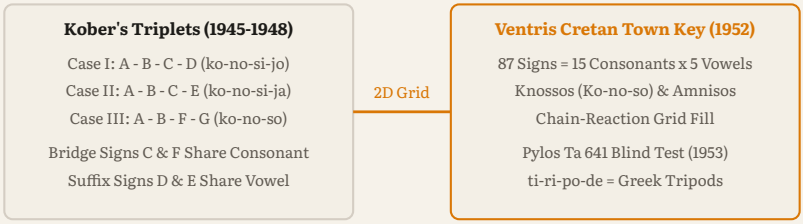


FIG. 6.1 — Alice Kober's inflectional triplets and Michael Ventris's consonant-vowel grid unlocking Mycenaean Greek.

Between 1943 and 1948, sorting her 180,000 scrap-paper cards by hand, Kober extracted the structural skeleton of the unknown language:

- **The Syllable Count Bottleneck**: Kober established that Linear B used roughly 87 active phonetic signs alongside commodity ideograms. Because an alphabetic script uses 20 to 30 signs and a logographic script uses thousands, a repertoire of 87 signs proved mathematically that Linear B was an open-syllable consonant-vowel (CV) syllabary combining five vowels with roughly fifteen consonants.
- **Kober's Triplets (Inflectional Paradigms)**: In a landmark 1945 paper, Kober proved that the language behind Linear B was heavily inflected, changing word endings to mark grammatical cases and genders. She isolated groups of three words (Kober's triplets) that shared the exact same root signs at the beginning (`A-B-...`) while varying their final signs across three recurring

grammatical patterns: Case I (A-B-C-D), Case II (A-B-C-E), and Case III (A-B-F).

- **The Bridge Syllable and the 2D Phonetic Grid**: In an open *CV* syllabary, when an inflected root ending in a consonant takes different vowel case-endings, the bridge syllable spanning the end of the stem and the start of the suffix must change its sign while keeping the same consonant. Thus Sign C (in Cases I and II) and Sign F (in Case III) shared the same consonant but carried different vowels. Chaining these intersecting triplets together in 1948, Kober built a two-dimensional grid of consonants and vowels without knowing a single sound value.

Kober died of cancer in May 1950. In London, British architect Michael Ventris expanded Kober's grid using Blegen's newly published Pylos tablets. In *Work Note 20* (June 1, 1952), Ventris noticed that five prominent inflectional triplets appeared on the Knossos tablets from Crete but were absent from Pylos on the mainland. They had to be Cretan towns (Case III) along with their masculine (Case I) and feminine (Case II) ethnic adjectives. Substituting the Cretan city names Amnisos (*A-mi-ni-so*), Knossos (*Ko-no-so*), and Phaistos (*Pa-i-to*) into Kober's grid triggered a chain reaction.

When Ventris read off the resulting words—*po-me* (*poimēn*, shepherd), *ka-ke-u* (*khalkeus*, bronzesmith), *ko-wo* (boy), and *ko-wa* (girl)—the language proved to be archaic Mycenaean Greek, written seven hundred years before Homer. Ten months after Ventris announced the decipherment on BBC Radio on July 1, 1952, with philologist John Chadwick, Carl Blegen dug up tablet Ta 641 at Pylos stamped with pictograms of three-legged cauldrons and four-handled goblets. Decoded with Ventris's grid, the syllables spelled out the Greek words *ti-ri-po-de* (tripods) and *qe-to-ro-we* (four-eared).

CHAPTER 7

The Burned Library of Leningrad

Yuri Knorozov and the Phonetic Key to Maya Hieroglyphs

Walking past a shelled library on the outskirts of Berlin in May 1945, Red Army artillery spotter Yuri Valentinovich Knorozov pulled two boxes from the ash. Inside lay two rare books that had survived unburned. The first was an 1864 Paris edition of Bishop Diego de Landa's sixteenth-century manuscript *Relación de las cosas de Yucatán*, edited by Charles Étienne Brasseur de Bourbourg. The second was Paul Schellhas's 1930 Berlin reproduction of the three surviving pre-Columbian Maya bark-paper books: the Dresden, Madrid, and Paris Codices. Knorozov carried both volumes back to Moscow.

Two years later, Knorozov asked his ethnography professor at Moscow State University, Sergei Tokarev, whether anyone could decipher the Maya script. Tokarev pointed to a 1945 article by Paul Schellhas arguing that the decipherment of Maya hieroglyphs was an insoluble problem. In the West, Maya epigraphy was ruled by British archaeologist J. Eric S. Thompson of the Carnegie Institution. Scholars since Ernst Förstemann in the 1880s could read the bar-and-dot numbers and Long Count calendar dates on Maya stelae. Yet the remaining eighty percent of the glyphs remained mute. Thompson enforced a strict orthodoxy. He maintained that the Classic Maya were peaceful astronomer-priests who never recorded wars, biographies, or spoken language on stone. According to Thompson, Maya glyphs were esoteric ideograms representing abstract concepts without spelling phonetic words.

Moving in 1949 to a tiny room inside the Museum of Ethnography in Leningrad, where he slept on a cot beside his desk, Knorozov

ignored Thompson's ban. He analyzed the three Maya codices using positional linguistics and sign-frequency statistics. Counting every distinct graphic element across the Dresden, Madrid, and Paris Codices while excluding compound ligatures and minor variants, he found that the Maya texts employed roughly 355 unique active signs.

That single number demolished Thompson's dogma. Every known writing system on Earth falls into one of three regimes based on its active sign inventory:

- **The Typological Sign-Count Law**: Purely alphabetic scripts (such as Latin or Cyrillic) represent phonemes and require 20 to 35 signs. Purely syllabic scripts (such as Linear B or Japanese kana) represent open syllables and require 50 to 100 signs. Mixed logosyllabic scripts (such as Egyptian hieroglyphs or cuneiform) combine logograms with phonetic syllable signs and employ several hundred to a few thousand core signs. With 355 active signs, Maya writing had to be a mixed logosyllabic script capable of spelling spoken Maya words phonetically.
- **Reinterpreting De Landa's Alphabet as Syllables**: In 1566, after burning Maya books at Maní, Bishop Diego de Landa forced a literate Maya noble, Gaspar Antonio Chi, to write down the glyphs matching the Spanish alphabet (`A, B, C...`). When nineteenth-century scholars plugged de Landa's twenty-seven glyphs into the codices as single letters, the results produced gibberish. Knorozov reconstructed the scene. When the bishop pronounced the Spanish letter names aloud, the scribe drew the Maya syllabic glyphs matching those spoken syllables (*a*, *be*, *se*, *ku*).
- **The Law of Synharmony (*CV*-*CV* Echo Spelling)**: Armed with de Landa's syllable signs, Knorozov turned to the Dresden Codex. Above a drawing of a turkey stood a two-glyph com-

pound. In colonial Yucatec dictionaries, a turkey was called *kutz*, a closed consonant-vowel-consonant (CVC) syllable. The first glyph was de Landa's sign for *ku* (the letter name Q). Because Maya phonetic signs represented open consonant-vowel (CV) syllables, Knorozov deduced **synharmony**. Maya scribes wrote a closed $C_1V_1C_2$ root using two open CV signs (C_1V_1-C_2V_1), where the second vowel echoed the first. Since turkey was *kutz*, the second sign had to be *tzu* (*ku*-*tzu*). Checking the glyph compound above a dog (*tzul* in Yucatec), Knorozov found that its first sign was the exact same *tzu* glyph, followed by de Landa's sign for *lu*, spelling *tzu*-*lu* (*tzul*, dog).

When Knorozov published his report in *Sovetskaya Etnografiya* in 1952 and defended his dissertation in March 1955, the committee awarded him an immediate doctorate. Thompson launched a twenty-year campaign to discredit him. Only after Thompson's death in 1975 did Western epigraphers led by Floyd Lounsbury, Linda Schele, Peter Mathews, and David Stuart test Knorozov's grid on the monuments of Palenque and Copán. Every glyph block obeyed Knorozov's logosyllabic rules. In 1990, Knorozov finally stood atop the temples of Tikal and Palenque, reading aloud the dynastic names of Classic Maya kings.

CHAPTER 8

The Compositor's Spelling

Charlton Hinman and the Optical Collator of the First Folio

Inside a basement vault at the Folger Shakespeare Library in Washington, D.C., two copies of William Shakespeare's 1623 First Folio lay side by side on iron cradles. Above them stood a six-foot steel frame rigged with surplus World War II reconnaissance lenses, mirrors, and rotating shutters. Sitting at the binocular eyepiece in 1949 was Johns Hopkins professor Charlton Hinman. During the war, as a U.S. Navy aerial intelligence analyst, Hinman had watched officers compare before-and-after photos of Pacific airfields. By superimposing both images through an optical blink-comparator that flashed the left and right pictures alternately, any newly dug trench or bunker flickered against the motionless background.

Returning to Renaissance scholarship, Hinman realized that the same optical physics could solve the hardest problem in English bibliography. He set out to reconstruct how William and Isaac Jaggard's London shop set and printed *Mr. William Shakespeares Comedies, Histories, & Tragedies* between February 1622 and November 1623. Eighteen of Shakespeare's thirty-six plays—including *Macbeth*, *The Tempest*, *Twelfth Night*, and *Julius Caesar*—survive exclusively because fellow actors John Heminge and Henry Condell gathered his manuscripts into the 1623 First Folio seven years after his death.

Before Hinman, editors assumed two premises about early modern printing. First, they assumed that every copy of the 1623 First Folio contained the exact same printed words. Second, they assumed that Jaggard's workmen had set the lead type for each

play sequentially from Page 1 to the end. Checking whether all copies were identical had been impossible. The First Folio contains roughly nine hundred double-column pages holding over 3.5 million pieces of hand-set type. Even after Henry Clay Folger amassed eighty-two of the roughly 235 surviving copies inside one building, comparing fifty copies letter by letter by naked eye would have required inspecting 175 million characters.

With Folger grants, Hinman built his custom optical collator and spent sixteen years—from 1947 to 1963—blinking fifty-five Folger copies of the First Folio against a reference copy at forty pages an hour. Whenever a pressman in 1623 had stopped the press to change a comma or fix a word, the altered character danced inside Hinman's eyepiece. Moreover, Hinman's lenses picked up microscopic scars on pieces of lead-antimony-tin type nicked on the shop floor. By tracking three streams of telemetry across fifty thousand folio pages, Hinman reconstructed Jaggard's printing room:

- **Stop-Press Variant Collation**: Whenever a proofreader spotted an error while a sheet was being printed, he pulled the faulty type from the iron chase and inserted a correction. Because imported rag paper constituted nearly half the cost of a book, the shop never discarded the faulty sheets already printed. Corrected and uncorrected sheets were stacked and bound at random, producing over five hundred stop-press variants across the fifty-five Folger copies.
- **Damaged Type Recurrence and Casting Off Copy**: By cataloging over six hundred scarred metal letters (such as a capital B with a broken loop) and charting every page on which those pieces reappeared after distribution back into the type cases, Hinman proved that Jaggard's shop owned too little type to set a play sequentially. Because each six-leaf gathering (quire)

consisted of three nested folio sheets folded inside one another, typesetters had to cast off copy. They estimated manuscript line counts ahead of time to set inner pages (Pages 6 and 7) first and outer pages (Pages 1 and 12) last. Whenever they miscalculated, they crowded two verses onto one line or padded scenes to hit the bottom of Page 6.

- **Compositor Spelling Fingerprints**: Early modern typesetters read a clause from the manuscript, held the words in working memory, and plucked letters from the type case using their own spelling habits. Cross-referencing spelling profiles (*doe/goe/heere* versus *do/go/here*) against the type cases tracked by damaged letters, Hinman isolated five workmen (Compositors A through E). Among them was Compositor E, a teenage apprentice named John Leason bound on November 4, 1622, who introduced hundreds of errors into *Romeo and Juliet*, *Hamlet*, *King Lear*, and *Othello*.

In 1963, Hinman published *The Printing and Proof-Reading of the First Folio of Shakespeare*, giving editors a reliability calibration for every column. Because Jaggard's pressmen corrected type on the fly and mixed sheets continuously during binding, across the fifty-five Folger copies Hinman collated, no two copies shared the exact same combination of corrected and uncorrected text across every play.

CHAPTER 9

The Bayesian Quill

Frederick Mosteller, David Wallace, and the Disputed Federalist Papers

Two authors writing on the same constitutional crisis in the same year, using the same eighteenth-century vocabulary and averaging 34.55 and 34.59 words per sentence respectively, cannot be told apart by traditional literary criticism. For 175 years, that stylistic convergence deadlocked American historians over the authorship of twelve essays inside *The Federalist*. Published in New York newspapers between October 1787 and August 1788 under the shared Roman pseudonym *Publius*, the eighty-five essays urged voters to ratify the United States Constitution.

Because the three authors—Alexander Hamilton, James Madison, and John Jay—kept their assignments secret during the ratification debates, no public attribution list existed until July 1804. Two days before crossing the Hudson River to Weehawken, New Jersey, for his fatal duel with Vice President Aaron Burr, Alexander Hamilton slipped a handwritten list into a bookshelf inside the law office of Judge Egbert Benson. In it he claimed personal authorship of sixty-three of the eighty-five *Federalist* numbers. A decade later, upon retiring from the presidency, James Madison published his own annotated list in the 1818 Jacob Gideon edition of *The Federalist*. Both men agreed that John Jay wrote five essays, Madison wrote fourteen undisputed essays, and Hamilton wrote fifty-one undisputed essays (plus three joint papers). However, Madison claimed sole authorship of twelve essays that Hamilton had claimed before his death: *Federalist* Numbers 49 through 58, and Numbers 62 and 63.

For a century and a half, historians tried to adjudicate the twelve disputed papers by hunting for political ideas or structural statistics. Every traditional test failed. Both Hamilton and Madison were writing from the same Philadelphia convention notes under tight deadlines. When statisticians measured sentence lengths across all undisputed papers, Hamilton averaged 34.55 words per sentence (standard deviation 19.2) while Madison averaged 34.59 words per sentence (standard deviation 20.3). Looking at political nouns like *executive*, *legislature*, *court*, *taxation*, or *war* also proved useless. The frequency of content words depends on the topic of an essay rather than who wrote it.

In 1959, encouraged by historian Douglass Adair—who had noticed that Hamilton habitually wrote *while* whereas Madison wrote *whilst*—Harvard statistician Frederick Mosteller and University of Chicago statistician David L. Wallace launched a three-year forensic investigation. Using Harvard's IBM 7090 mainframe computer and student assistants punching word cards by hand, Mosteller and Wallace realized that a writer's true fingerprint lies in low-level, context-free **function words**. Prepositions, conjunctions, articles, and common modifiers (*by*, *from*, *to*, *upon*, *whilst*, *enough*, *also*, *an*) keep nearly invariant usage rates regardless of whether an essay discusses naval militias, judicial tenure, or Greek confederacies.

Screening 165 candidate markers across forty-eight known Hamilton essays (94,000 words) and fifty known Madison essays and external pamphlets (114,000 words), Mosteller and Wallace isolated thirty discriminator words. The sharpest separator was the preposition **upon**. Across forty-eight undisputed essays, Hamilton used *upon* an average of 3.24 times per thousand words and never wrote a thousand-word essay without it. Madison used *upon* only 0.23 times per thousand words, omitting it completely in over eighty

percent of his papers. Similarly, Madison used *by* 11.4 times per thousand words against Hamilton's 7.3, while Hamilton used *to* 40.8 times per thousand words against Madison's 35.2.

Because writers repeat a word slightly more often once it enters short-term memory, word counts in real prose are mildly overdispersed compared to a memoryless Poisson process. Mosteller and Wallace modeled the probability density $f_M(x_i)$ and $f_H(x_i)$ of observing count x_i of word i under a negative-binomial distribution. They combined all $k = 30$ independent function words $\mathbf{x} = (x_1, x_2, \ldots, x_k)$ via Bayes' theorem to compute the log posterior odds between Madison (M) and Hamilton (H):

$$\ln \frac{P(M \mid \mathbf{x})}{P(H \mid \mathbf{x})} = \ln \frac{P(M)}{P(H)} + \sum_{i=1}^{k} \ln \frac{f_M(x_i)}{f_H(x_i)}$$

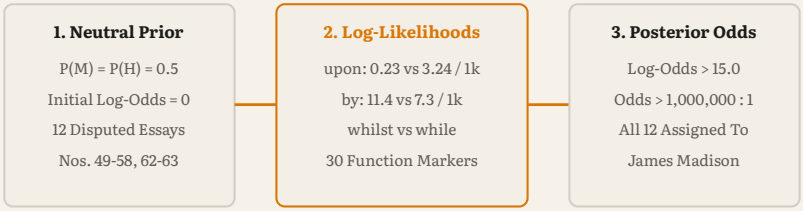


FIG. 9.1 — Mosteller and Wallace's Bayesian log-odds update across 30 negative-binomial function words resolving the 12 disputed Federalist essays.

Even when Mosteller and Wallace set the initial prior odds $\frac{P(M)}{P(H)}$ to 1 or handicapped Madison by ten thousand to one, the cumulative log-likelihood sum overwhelmed the prior. In every one of the twelve disputed essays, *upon* was virtually absent, *whilst* appeared repeatedly, and the prepositional ratios aligned with Madison's

negative-binomial baseline. Only *Federalist* No. 55 showed slightly weaker odds (roughly 90 to 1 for Madison). Across the remaining eleven disputed essays, Mosteller and Wallace’s likelihood ratios favored James Madison by odds exceeding millions to one, turning statistical stylometry into a standard forensic tool.

CHAPTER 10

The Sunken Papyrus

Bernard Grenfell, Arthur Hunt, and the Trash Heaps of Oxyrhynchus

Outside the village of Behnesa on the western rim of the Nile Valley, one hundred miles south of Cairo, thirty-foot-high mounds of ancient garbage stretched for over a mile along the desert edge. In December 1896, Oxford classical scholars Bernard Pyne Grenfell and Arthur Surridge Hunt rode up to the site on donkeys. Both fellows of The Queen's College, Oxford, had been sent by the Egypt Exploration Fund not to hunt for royal tombs, but to dig through the rubbish dumps of the Greco-Roman provincial capital of Oxyrhynchus, named after the sharp-nosed fish sacred to the district.

Over the previous two centuries, classical scholars had confronted a grim reality: more than ninety-five percent of ancient Greek and Roman literature had perished. Of the 120 plays written by Sophocles, only seven survived the Middle Ages. Of Euripides' ninety plays, only nineteen remained. Of the nine lyric books of Sappho admired across antiquity, only one complete poem had been copied into medieval parchment codices. Across Italy, Greece, and Asia Minor, the standard writing paper of the ancient Mediterranean—sheets made from the sliced pith of the marsh reed *Cyperus papyrus*—rotted within decades whenever exposed to rain or soil humidity.

Even in rainless Egypt, Grenfell and Hunt realized that traditional archaeologists were digging in the wrong microclimate. Stone temples, palaces, and tombs inside the Nile floodplain sat below the annual percolation level of the Nile inundation. Rising groundwater soaked the foundations and turned stored papyrus

rolls into mud. Looking across the desert edge behind Behnesa in January 1897, Grenfell and Hunt deduced where the dry paper had gone. For a thousand years—from Alexander the Great's conquest in 332 BCE until the Arab conquest in 641 CE—the Greek-speaking administrators, teachers, merchants, and Christians of Oxyrhynchus had cleaned out their bookcases and filing offices. Whenever papyrus rolls grew frayed or tax ledgers expired, servants carried baskets of torn paper outside the walls and dumped them onto trash mounds (called *koms* in Arabic) well above the Nile damp line, where desert sand buried and sealed each layer.

After digging test trenches in the cemetery for three weeks, Grenfell and Hunt shifted their seventy workmen to a low garbage mound on January 11, 1897. Within hours, a workman handed Hunt a crumpled leaf of third-century CE papyrus written on both sides in Greek uncials. Brushing off the dirt, Hunt read the header *IESOU LOGIA* (Sayings of Jesus), a collection of unknown sayings of Jesus matching the lost *Gospel of Thomas* fifty years before the Nag Hammadi discovery. The next day, they pulled out the oldest surviving fragment of the Gospel of Matthew (*P.Oxy.* 1 and *P.Oxy.* 2).

Across six winter seasons between 1897 and 1907, employing up to two hundred diggers, Grenfell and Hunt trenched through the Oxyrhynchus *koms* down to thirty feet until damp subsoil appeared. Papyri poured out so fast that the two scholars sat inside their tent until midnight damping twisted rolls between wet towels and packing them into tin biscuit boxes:

- **The Stratigraphy of the* Afsh *Layer: Experienced diggers learned to spot the subterranean soil texture that preserved papyrus—a porous mixture of dry sand, decayed palm leaves, ash, and potsherds called* afsh. *Wherever a stratum of* afsh* sat capped beneath a hard crust of upper sand above the water table, wads of

third-century library rolls emerged so crisp they crackled when unfolded.

- **Vanished Classical Masterpieces**: From the baskets of torn book-rolls came four hundred lines of Sophocles' lost satyr play *Ichneutae* (*The Trackers*), extensive scenes from Menander's comedies, lost poems of Sappho and Pindar, a lost epitome of Livy, and the *Hellenica Oxyrhynchia*, an unknown fourth-century BCE Greek history continuing Thucydides.
- **The Unfiltered Documentary Archive**: Ninety percent of the haul consisted of everyday administrative and private paperwork. The mounds yielded horoscopes, wet-nurse contracts, gladiatorial programs, tax receipts, and private correspondence. Among them was a petulant letter from a Greek schoolboy named Theon threatening to refuse to write, speak, or wish health to his father if he was not taken along to Alexandria.

Stored in seven hundred biscuit boxes at Oxford's Sackler Library, the five hundred thousand fragments from Oxyrhynchus have restored lost poems of Sappho, plays of Sophocles, early Christian gospels, and seventy percent of all surviving literary papyri.

CHAPTER 11

The Iron-Gall Prayer Book

Reviel Netz, William Noel, and the X-Ray Archimedes Palimpsest

Cataloged for centuries as *Euchologion* 329, a burnt, mold-spotted thirteenth-century Byzantine prayer book of 174 goatskin leaves sat on an auction block at Christie's in New York on October 29, 1998. When the gavel fell at two million dollars to an anonymous American technology billionaire, the buyer placed the volume into the hands of curator William Noel at the Walters Art Museum in Baltimore for open scientific study. Researchers soon confronted the most battered crime scene in the history of mathematics.

Ninety-two years earlier, in the summer of 1906, Danish philologist Johan Ludvig Heiberg had traveled to Constantinople to inspect that same prayer book inside the Metochion of the Holy Sepulchre. He had noticed a brief catalog entry by paleographer Athanasios Papadopoulos-Kerameus mentioning mathematical diagrams beneath the prayers. Examining the pages with a magnifying glass, Heiberg realized that the prayer book was a **palimpsest** (from the Greek *palin psao*, scraped again). Because parchment was expensive in the medieval Mediterranean, a Christian scribe named Johannes Myronas—who dated his colophon on April 14, 1229, almost certainly in Jerusalem—had recycled seven older Greek manuscripts. He washed and scraped their ink off with weak acid and pumice, cut the sheets in half down their fold, rotated them ninety degrees, and copied liturgical prayers across the faint brown ghosts of the older writing.

The underlying tenth-century manuscript, copied in Constantinople around 975 CE, contained seven treatises by Archimedes of

Syracuse (287–212 BCE). Even more critically, this scraped codex held the only surviving Greek text of *On Floating Bodies* and the only copies anywhere on Earth of two works long thought lost: *Stomachion* (a treatise on combinatorics) and *The Method of Mechanical Theorems*. Written as a letter from Archimedes to Eratosthenes at Alexandria, *The Method* explained how Archimedes discovered geometric theorems by weighing infinitely thin slices on an imaginary balance.

After Heiberg photographed a fraction of the pages in 1906 and 1908, the manuscript vanished after World War I and remained hidden in a French private collection for seventy years. By the time William Noel and Stanford historian Reviel Netz opened the codex in Baltimore in 1999, its condition was catastrophic. Thick mold had perforated the parchment, and modern glue bound the spine too tightly to photograph the inner margins. Worst of all, after 1929 an unscrupulous owner had tried to inflate the book's value by painting four Byzantine-style gold-leaf and lead-white Evangelist icons directly on top of four pages of Archimedes' text.

Working with imaging scientists Roger Easton Jr., Keith Knox, and Bill Christens-Barry, the team conserved and separated every leaf. They imaged the exposed pages under multispectral light, flashing ultraviolet (365 nm) and red (625 nm) diodes so the brown iron-gall ink of Archimedes vanished in the red channel while the black prayer text remained. That contrast allowed digital subtraction of the prayers. Yet multispectral light could not penetrate the solid gold leaf and lead paint of the four forged icons.

In 2004, physicist Uwe Bergmann at the Stanford Linear Accelerator Center (SLAC) proposed a nuclear-physics solution. Both the tenth-century scribe and the thirteenth-century priest had written with **iron-gall ink**, brewed by mixing crushed oak galls (tannic acid) with ferrous sulfate ($\mathrm{FeSO_4}$). Even when Myronas

scraped away the surface crust of the tenth-century letters in 1229, trace iron atoms remained chelated deep inside the collagen fibers. At the Stanford Synchrotron Radiation Lightsource (SSRL), Bergmann aimed a 50-micron beam of synchrotron X-rays tuned to $7.11\ \mathrm{keV}$—just above the K-shell binding energy of iron—directly through the gold paintings. As the X-ray beam penetrated thickness x with attenuation coefficient $\mu(E)$, every buried iron atom inside Archimedes' inkstrokes ejected an inner-shell electron and emitted X-ray fluorescence photons at iron's $K\alpha$ spectral line:

$$I(x) = I_0 \exp(-\mu(E) x), \quad E_{K\alpha}(\mathrm{Fe}) = 6.40\ \mathrm{keV}$$

Because $6.40\ \mathrm{keV}$ X-rays pass back out through gold leaf and parchment—and enabled reading both sides of a leaf using front and back detectors—the synchrotron mapped the iron atoms at 40-micron resolution. The scans also uncovered ten pages of lost speeches by the Athenian orator Hyperides. As the scan lines assembled in August 2006, the forged gold icons vanished and revealed Archimedes calculating the volume of solids with actual infinite sets nineteen centuries before Newton and Leibniz.

CHAPTER 12

The Carbonized Scrolls

Brent Seales and the Virtual Unwrapping of the Herculaneum Library

Carbon ink written on a carbonized papyrus scroll possesses the exact same atomic number—$Z = 6$—as the burnt reed fibers underneath it, rendering two thousand years of classical literature invisible to conventional X-ray radiography. For twenty-five years, that law of atomic physics blocked every attempt to read the last intact library surviving from Greco-Roman antiquity: the charred scrolls of Herculaneum.

In the autumn of 79 CE, when Mount Vesuvius erupted above the Bay of Naples, Herculaneum was struck by superheated pyroclastic surges. Avalanches of volcanic gas and ash traveled at over a hundred kilometers per hour at temperatures reaching 320 degrees Celsius. Inside the Villa of the Papyri, owned by Julius Caesar's father-in-law Lucius Calpurnius Piso Caesoninus, the blast wave flash-heated a philosophical library of roughly 1,800 Greek and Latin papyrus scrolls. Twenty meters of volcanic mud then smothered the rooms and hardened into tufa rock. Deprived of oxygen, the scrolls did not burn to ash. Instead, like wood heated in a charcoal kiln, the cellulose of the papyrus reed underwent pyrolysis, transforming every scroll into a black cylinder of carbon.

Rediscovered between 1752 and 1754 when Swiss engineer Karl Weber and workmen dug shafts sixty feet through the volcanic rock, the Herculaneum scrolls proved agonizingly fragile. Early attempts to slice the rolls open in Naples destroyed hundreds of columns. Even after Vatican scriptor Father Antonio Piaggio invented a traction frame in 1756 that glued ox-gut membrane to a

scroll and peeled back layers with silk threads, hundreds of tighter scrolls remained fused. Whenever mechanical tension touched the brittle charcoal wafers, they shattered into black flakes. Curators in Naples, Paris, and London eventually locked roughly six hundred unopened scrolls away.

In 2015, University of Kentucky computer scientist W. Brent Seales proved that a burned scroll could be read without opening it when his team virtually unwrapped the carbonized sixth-century CE *En-Gedi Scroll* from Israel, discovering the Book of Leviticus inside a lump of charcoal. Yet the En-Gedi ink contained trace lead and iron salts that glowed brightly in a micro-CT scanner. First-century Roman and Greek scribes, however, wrote on papyrus using *atramentum librarium*, an ink ground from pine-pitch lampblack soot mixed with gum arabic. Because lampblack soot contains no heavy metals, firing X-rays at a Herculaneum scroll produced zero absorption contrast between the ink and the page.

Refusing to concede that the ink had left no trace, Seales carried two unopened scrolls from the Institut de France in Paris (*PHerc. Paris. 3* and *PHerc. Paris. 4*) and detached fragments to the Diamond Light Source synchrotron in Oxfordshire in 2019. There he scanned them with phase-contrast micro-computed tomography at a voxel resolution of 8 micrometers ($0.008\ \mathrm{mm}$). When entrepreneurs Nat Friedman and Daniel Gross joined Seales in March 2023 to launch the $1,000,000 **Vesuvius Challenge**, the global team built a three-stage pipeline to solve the zero-contrast bottleneck:

- **8-Micron Synchrotron Phase-Contrast Micro-CT**: Instead of measuring only X-ray absorption (which depends on atomic number Z), synchrotron phase-contrast tomography measures the refractive phase shift of coherent X-ray wavefronts passing across microscopic density boundaries. That phase shift re-

solves every cellular vascular bundle of the papyrus pith inside the sealed scroll at 8-micron 3D voxel resolution.

- **3D Topological Mesh Segmentation and Virtual Flattening**: Because the volcanic mud had crushed the spiral rolls into twisted accordions with over a hundred overlapping wraps, competitors built automated segmentation algorithms (*ThaumatoAnakalyptor*). These traced a continuous manifold surface through the 3D point cloud and unrolled each virtual sheet into a flat plane without crossing adjacent layers.
- **Sub-Micron Ink Relief Detection via 3D Neural Networks**: Unlike fluid iron-gall ink that soaks into parchment, viscous carbon-soot ink sat as a dried crust on top of the papyrus fibers, creating a sub-micron thickness bump and a crackled surface texture (*craquelure*) when carbonized. Training 3D neural networks on infrared photographs of broken fragments, student Luke Farritor detected the first complete Greek word inside the unopened Paris scroll in October 2023: ΠΟΡΦΥΡΑΣ (*porphyras*, purple).

Building on that breakthrough, Youssef Nader, Luke Farritor, and Julian Schilliger combined autonomous sheet segmentation with ensemble 3D ink-detection transformers to capture the Grand Prize. In February 2024, without unrolling a millimeter of the charcoal cylinder, the team extracted fifteen columns of Greek text by the Epicurean philosopher Philodemus on music, food, and the mechanics of human pleasure.